BA820 – Project M2

Cover Page

- **Project Title:** Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 - Team 10
- **Student Name:** Shanmathi Sivakumar

1. Refined Problem Statement & Focus (~0.5 page)

My project examines whether age gaps in films with multiple romantic couples tend to concentrate in a single standout pairing ("headline" pattern) or recur across multiple couples within the same film ("recurrence" pattern). This distinction matters because a headline pattern is more consistent with age gaps being used as a deliberate narrative device, while recurrence suggests a broader within-film casting norm.

Initially, my framing assumed I could compare the "main couple" vs "side couples" using the dataset's couple_number field. However, my EDA showed that this assumption does not hold. The dataset is structured so that actor_1 is always the older (or same-age) partner, meaning couples are recorded in a consistent age-order rather than by narrative importance. In multi-couple films, couple_number appears to function mainly as an ordering/indexing variable (often aligning with larger age gaps) rather than a true indicator of the main relationship. As a result, I revised the framing: in each film, the "headline" couple is defined as the couple with the largest age gap, and recurrence is measured by whether the second largest (and additional) age gaps are also large (e.g., ≥ 20 years). This updated definition better matches the structure of the dataset and supports a more valid within-film comparison.

2. EDA & Preprocessing: Updates (~0.75 page)

In M1, my EDA was mainly focused on understanding the dataset at a basic level and checking whether it is usable for studying age gaps in movie couples. I looked at the overall distribution of age gaps, how it changes across different types of pairings, and how it behaves over time. A key motivation that came out of M1 is that the age gap distribution is not symmetric. Most couples have smaller gaps, but there is a noticeable long tail of movies with much larger gaps. That made me interested in where those large gaps are showing up and whether they appear as isolated cases or as part of a pattern inside the same film. M1 helped confirm that the dataset is clean and structured enough to support deeper analysis without needing major fixes.

In M2, I added new EDA specifically designed to answer my personal question about within film concentration versus recurrence. The biggest update was shifting the unit of analysis from individual couples to movies with multiple couples, because my question is about patterns within the same movie. I first checked how common multi couple movies are and found that there is a decent subset of movies where within film comparisons make sense, but most of those movies only have two couples. That mattered because it changed how I define recurrence for the typical case. I also challenged one of my original assumptions about the dataset. I initially thought couple number 1 could represent the main couple, but after checking the structure, actor 1 is basically always the older partner and couple numbering does not clearly represent story importance. Because of that, I stopped using couple number as a main couple indicator and instead defined the headline couple in a measurable way as the couple with the largest age gap in that film.

For preprocessing, I did not need heavy cleaning because there were no major missing value issues in the columns I used. The main preprocessing updates in M2 were feature building steps to support the within film analysis. I created movie level summary variables like the second largest age gap in the movie and a dominance measure that compares the largest gap to the second largest. I also used a simple cutoff of 20 years to flag large gaps, which makes it easier to talk about recurrence in plain language. These updates were necessary because my question is not just about how big gaps are overall, but whether they repeat across couples inside the same movie.

3. Analysis & Experiments (~1.5 page)

For this milestone, I wanted to move past just describing age gaps and test whether multi couple movies fall into different "types" based on my main idea: do age gaps concentrate in one standout pairing, or do they show up across multiple couples in the same movie. Since there are no labels for "headline" vs "recurrence" movies, clustering felt like a good fit because it lets the patterns come from the data.

Method 1: K Means clustering on movie level features:

Because my question is about patterns within a film, I switched from the couple level table to a movie level dataset. I also focused only on multi couple movies, since single couple films can't tell me anything about within movie recurrence. That left me with 224 movies. I built a feature set that maps directly to my question. I used top gap dominance, second gap, median gap, count of large gaps, and number of couples. Together, these capture whether the biggest gap is way larger than the rest, whether the second biggest gap is also large, what a "typical" gap looks like in that film, and how many chances the movie even has for recurrence. I scaled the features because K Means is distance based and I didn't want one column to dominate just because it has a bigger numeric range.

The main decision with K Means is choosing k. I tested k from 2 to 8 and tracked inertia (elbow) and silhouette score. The elbow kept dropping as k increased, but the improvements started slowing down. Silhouette was highest at k=2, but that felt too simple and kind of forced everything into one split. I ended up using k=3 as a middle ground because it still had decent separation while giving me clusters that were big enough to interpret. After fitting k=3, I summarized each cluster using medians of the original features. That part was important because at first the clusters are just numbers. Looking at the medians helped me give them "story" labels, like headline dominant, recurrence leaning, and low gap typical. I'm not saying these are perfect categories, but it makes the results easier to explain in terms of my question.

One thing I struggled with at first was expecting clustering to automatically explain itself. It doesn't. The cluster profile table was what made it actually useful, because it showed what each group looks like in plain terms.

Method 2: PCA as an experiment to improve clustering quality

At first I used PCA only as a visualization step, since it lets me plot the clusters in 2D even though my feature space is 5D. But after seeing the scree plot, I realized PCA could also be used as an experiment to improve clustering, not just for plotting. The scree plot showed that the first three components capture about 94 percent of the variance, so I tried running K Means in PCA space and compared the results to clustering in the original feature space. I tested PCA with 2 to 5 components and checked silhouette and inertia.

This was honestly the most useful surprise. Clustering on the original five features had silhouette around 0.38. PCA 2 components improved it to about 0.43. PCA 3 components improved it again to about 0.48, which was the best. But once I went to PCA 4 or 5 components, the silhouette dropped back down near the original result. That made sense once I thought about it because PCA 5 is basically the full space again, so it doesn't really remove noise or redundancy. Because of that, the version that worked best was K Means on the first 3 principal components. It gave the cleanest separation while still keeping most of the information. The downside is PCA components are harder to interpret, so I still explain the clusters using the original feature medians, since that ties back to the actual business story.

Overall, what I learned is that it's not enough to run one unsupervised method once. I needed to try different versions, check if the clusters actually look meaningful, and then connect the results back to the question I care about. PCA ended up being a helpful adjustment because it improved the cluster quality and gave me a stronger justification for the final setup.

4. Findings & Interpretations (~0.75 page)

From the EDA, the first important thing I found is that multi-couple movies are not the majority, but they are still a meaningful part of the dataset. About 27 percent of movies have 2 or more couples, so within-movie comparisons are possible. But the distribution is very skewed, because most of those multi-couple movies only have two couples. That matters a lot for my question, since "recurrence" in a two-couple movie is basically an all-or-nothing situation: either both couples have large age gaps or they do not, and that ends up being rare. Another key finding is that I cannot treat the couple numbering as "main couple vs side couple." The dataset is structured so actor 1 is almost always the older partner, and couple numbers seem to be more of an ordering system than a story-importance system. So instead of focusing on couple one as the "headline romance," I used a more reliable comparison: the largest age gap in a movie versus the second-largest gap.

When I tested the standout versus recurrence idea directly, the results leaned strongly toward a standout pattern. In multi-couple movies, it is uncommon for the second-largest age gap to also be large—only around 4 to 5 percent of multi-couple movies have a second-largest gap that is 20 years or more. This means that in most cases, if there is a large age gap in a movie, it tends to be concentrated in one pairing rather than showing up across multiple couples. At the same time, movie structure changes the story slightly: recurrence was basically non-existent for two-couple movies, but it became more noticeable for movies with three or more couples. For that group, about 13 percent of movies had at least two large-gap couples, suggesting recurrence is more likely in ensemble-style movies where the same casting pattern has more chances to appear.

The clustering analysis reinforced that multi-couple movies are not all doing the same thing. Using K-means, movies separated into clear types based on features tied to my question: a headline-dominant group (largest gap stands out), a recurrence-leaning group (higher second gaps and more large-gap couples), and a lower-gap/typical group (smaller gaps overall). PCA supported this structure as well—using three principal components reduced overlap and made the cluster separation clearer. Overall, these findings suggest large age gaps are usually used as a standout narrative element rather than repeated across all couples in the same film, though recurrence becomes more plausible in the smaller set of movies with three or more couples, which may point to stronger within-film casting norms in those productions.

5. Next Steps (~0.25 page)

- What I have not done yet is explain what kinds of movies are driving the recurrence cases. I also have not checked whether these patterns change over time or by couple type.
- Next, I plan to break the multi couple movies into groups by decade and pairing type and then rerun the same recurrence measures to see if the standout vs recurrence pattern is stable. I also want to test sensitivity to the large gap cutoff by trying a second threshold like 15 years, since recurrence at 20 years is extremely rare in two couple movies.
- The main unresolved question is what makes the “recurrence leaning” movies different in a real-world sense. A new question that came up is whether the rarity of recurrence is mainly caused by the cutoff definition rather than an actual casting pattern.
- These next steps are justified because my current results show strong evidence of a standout pattern overall, but also show recurrence becomes more plausible in three plus couple movies, so the natural follow up is to figure out when and why those recurrence cases happen.

Appendix

Shared GitHub Repository (Required)

- **Shared team GitHub repository: Shared GitHub**
- **Individual work folder: Individual folder**
- **Colab link: Notebook**

Supplemental Material (Highly Recommended)

Figure 1:

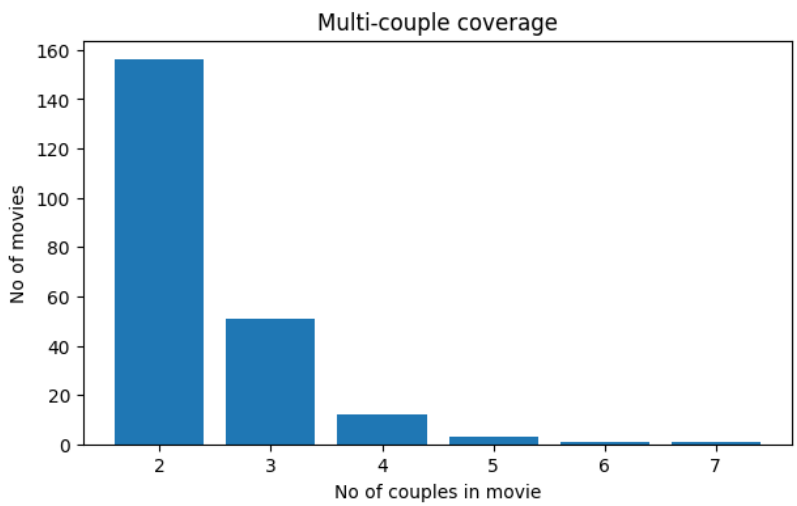

Figure 2: Couple Ordering Validity

```
Overall: actor_1_age >= actor_2_age in 100.00% of rows
Overall: actor_1_age >  actor_2_age in 97.40% of rows
Couple #1: actor_1_age >= actor_2_age in 100.00% of rows
Couple #1: actor_1_age >  actor_2_age in 98.55% of rows
```

Figure 3: Recurrence Rate by Movie Structure (2 Couples vs 3+ Couples)

	bucket	n_movies	recurrence_rate
0	2 couples	156	0.6
1	3+ couples	68	13.2

Figure 4:

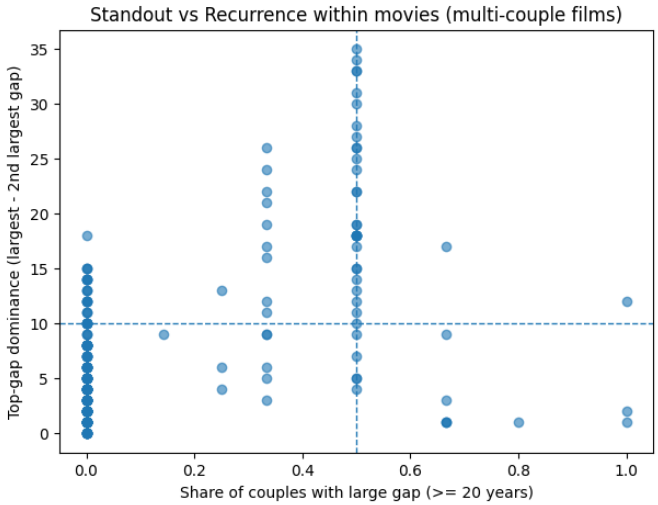

Figure 5:

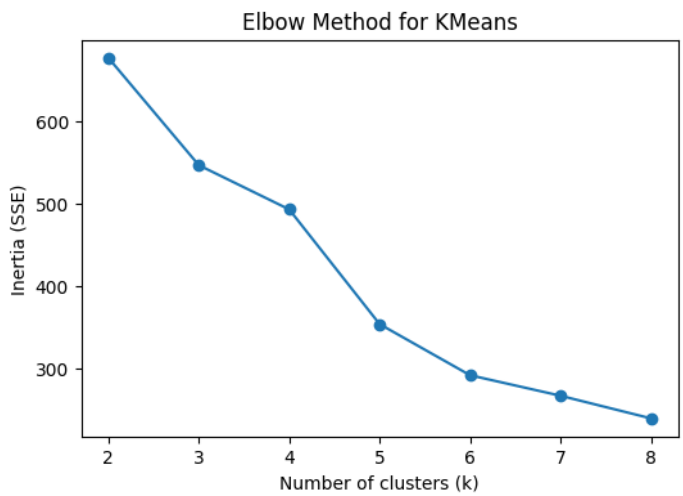

Figure 6: k-mean and silhouette result table

k	inertia	silhouette
2	676.833175	0.471301
3	546.650680	0.379884
4	492.983435	0.278306
5	353.828095	0.323907
6	291.854230	0.331321
7	266.979992	0.328847
8	239.503850	0.320168

Figure 7:

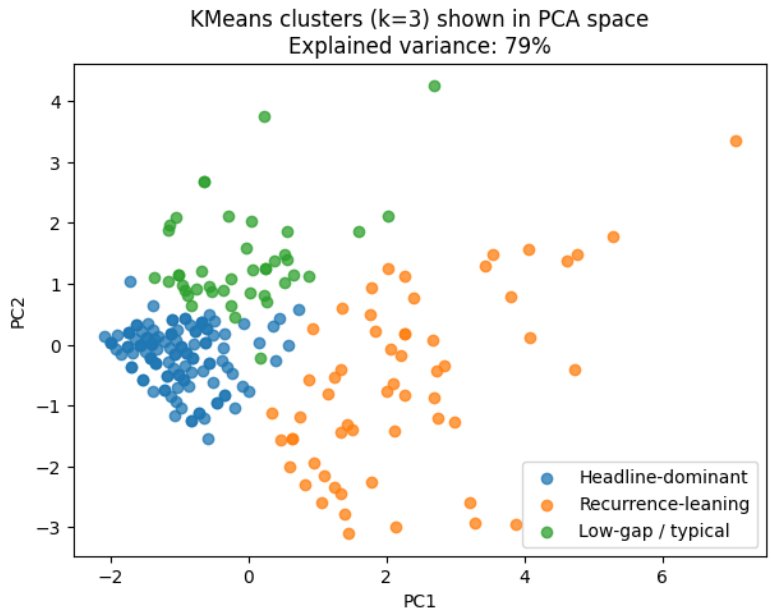

Figure 8:

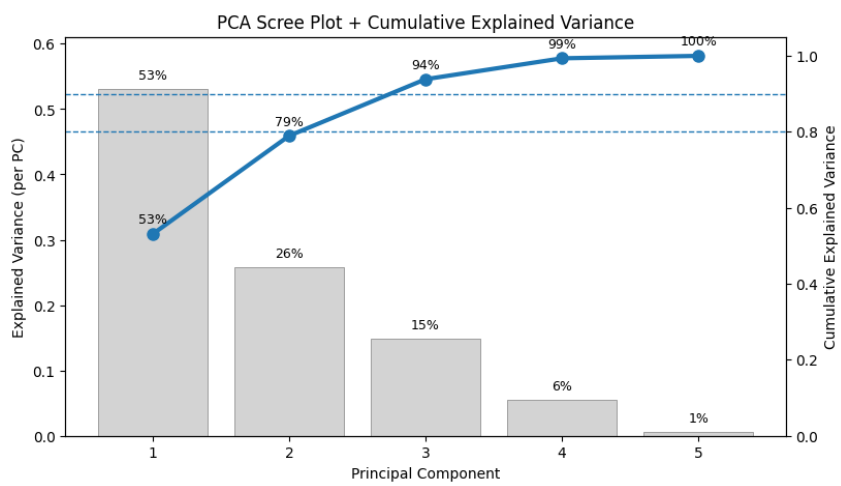

Figure 9:

space	inertia	silhouette
Original 5 features	546.650680	0.379884
PCA 2 PCs	346.554801	0.429839
PCA 3 PCs	464.939110	0.480166
PCA 4 PCs	539.517951	0.381096
PCA 5 PCs	546.650680	0.379884

Process Overview

I started by loading the age gap dataset from GitHub into Colab and doing quick checks to confirm the structure and data types. Then I created a few simple features that I needed for my question, like age gap, a large gap flag (20+ years), and a movie level key using movie name and release year. Since my question is about patterns within the same film, I shifted from couple level rows to a movie level table and focused only on multi couple movies (2+ couples). After that, I did question focused EDA to understand how common multi couple films are, confirm that couple ordering is not the "main couple," and measure whether large gaps are concentrated in one couple or repeat across couples, including comparing 2 couple vs 3+ couple movies. Finally, I ran unsupervised experiments using K means clustering and used PCA both for visualization and as a dimensionality reduction test to see whether clusters become cleaner, then I summarized everything in non-technical terms to connect back to the standout vs recurrence story.

Use of Generative AI Tools

I used ChatGPT as a supportive tool to help with idea generation, drafting text, and clarifying technical or conceptual questions during the project. All suggestions provided by the AI were critically evaluated, refined, and adapted to fit our own approach and understanding. The final work represents my own analysis, reasoning, and contributions, and AI was used only as an assistant, not as a replacement for my thinking.

Documentation of AI Usage:

Link to Chatbot: https://chatgpt.com/share/698a9c78-b844-800f-a25c-97dd58544c4f